

CGPA Calculation Using C

Programming Laboratory Report

Aim

To develop a C program that accepts semester-wise course details, validates all inputs, converts letter grades into grade points, calculates SGPA for each semester, and finally calculates the cumulative CGPA.

Variables and Data Types

Variable(s)	Data Type	Purpose
global_id, m, n, semester, course, code, credit, GP, semester_TC, semester_TGC, cumulative_TC, cumulative_TGC	int	IDs, counts, course data, grade points, and credit totals
grade	char	Stores the letter grade (O, A, B, C, D, E, or P)
SGPA, CGPA	float	Stores semester GPA and cumulative GPA

Algorithm

1. Start.
2. Define the variables and their types: global_id, m, n, semester, course, code, credit, GP, semester_TC, semester_TGC, cumulative_TC, and cumulative_TGC as integers; grade as a character; and SGPA and CGPA as floating-point variables.
3. Get global_id.
4. Validate global_id: If global_id is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
5. Get m.
6. Validate m: If m is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
7. Set the cumulative values: cumulative_TC = 0 and cumulative_TGC = 0.
8. Repeat the following for m semesters:
 - 8.1 Set semester_TC = 0 and semester_TGC = 0.
 - 8.2 Get n.
 - 8.3 Validate n; repeat input until n is positive.
 - 8.4 Repeat for n courses:
 - 8.4.1 Get code.
 - 8.4.2 Validate code; repeat until code is positive.
 - 8.4.3 Get credit.
 - 8.4.4 Validate credit; repeat until credit is positive.
 - 8.4.5 Get grade.
 - 8.4.6 Validate grade: accept only O, A, B, C, D, E, or P; otherwise request it again.
 - 8.4.7 Find GP using: O = 10, A = 9, B = 8, C = 7, D = 6, E = 5, P = 4.

- 8.4.8 semester_TGC = semester_TGC + (GP * credit).
- 8.4.9 semester_TC = semester_TC + credit.
- 8.5 SGPA = semester_TGC / semester_TC.
- 8.6 Display semester and SGPA.
- 8.7 cumulative_TGC = cumulative_TGC + semester_TGC.
- 8.8 cumulative_TC = cumulative_TC + semester_TC.
9. Calculate CGPA: CGPA = cumulative_TGC / cumulative_TC.
10. Display global_id, m, cumulative_TC, and CGPA.
11. Stop.

C Program

```
#include <stdio.h>
#include <ctype.h>

int main()
{
    int global_id, m, n, semester, course;
    int code, credit, GP;
    int semester_TC, semester_TGC;
    int cumulative_TC = 0, cumulative_TGC = 0;
    char grade;
    float SGPA, CGPA;

    do {
        printf("Enter Global ID: ");
        scanf("%d", &global_id);
        if (global_id <= 0) printf("Invalid Global ID. Enter a positive value.\n");
    } while (global_id <= 0);

    do {
        printf("Enter number of semesters: ");
        scanf("%d", &m);
        if (m <= 0) printf("Invalid number of semesters. Enter a positive value.\n");
    } while (m <= 0);

    for (semester = 1; semester <= m; semester++) {
        semester_TC = 0;
        semester_TGC = 0;
        do {
            printf("\nEnter number of courses in semester %d: ", semester);
            scanf("%d", &n);
            if (n <= 0) printf("Invalid number of courses. Enter a positive value.\n");
        } while (n <= 0);

        for (course = 1; course <= n; course++) {
            do {
                printf("Enter course code for course %d: ", course);
                scanf("%d", &code);
                if (code <= 0) printf("Invalid course code. Enter a positive value.\n");
            } while (code <= 0);

            do {
                printf("Enter credit: ");
                scanf("%d", &credit);
                if (credit <= 0) printf("Invalid credit. Enter a positive value.\n");
            } while (credit <= 0);
```

```

do {
    printf("Enter grade (O/A/B/C/D/E/P): ");
    scanf(" %c", &grade);
    grade = toupper((unsigned char)grade);
    if (grade!='O' && grade!='A' && grade!='B' && grade!='C' &&
        grade!='D' && grade!='E' && grade!='P')
        printf("Invalid grade. Enter O, A, B, C, D, E or P.\n");
} while (grade!='O' && grade!='A' && grade!='B' && grade!='C' &&
        grade!='D' && grade!='E' && grade!='P');

switch (grade) {
    case 'O': GP = 10; break;
    case 'A': GP = 9; break;
    case 'B': GP = 8; break;
    case 'C': GP = 7; break;
    case 'D': GP = 6; break;
    case 'E': GP = 5; break;
    default: GP = 4;
}
semester_TGC += GP * credit;
semester_TC += credit;
}

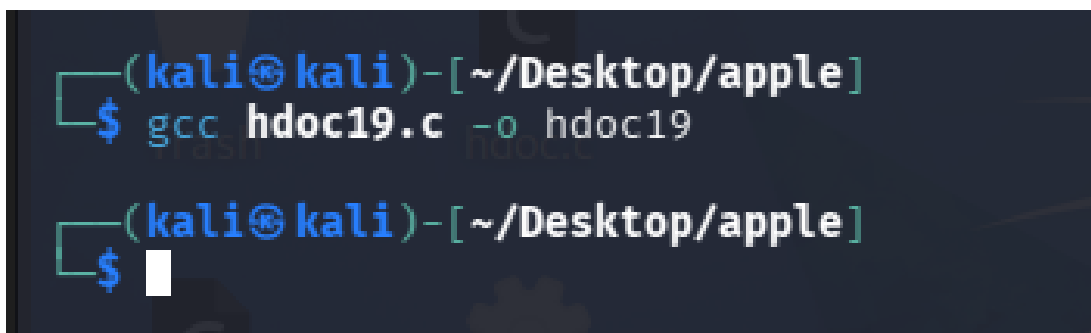
SGPA = (float)semester_TGC / semester_TC;
printf("Semester %d SGPA = %.2f\n", semester, SGPA);
cumulative_TGC += semester_TGC;
cumulative_TC += semester_TC;
}

CGPA = (float)cumulative_TGC / cumulative_TC;
printf("\nGlobal ID = %d\n", global_id);
printf("Number of Semesters = %d\n", m);
printf("Cumulative Credits = %d\n", cumulative_TC);
printf("CGPA = %.2f\n", CGPA);
return 0;
}

```

Program Output / Screenshots

Screenshot 1 - Compilation



```

(kali@kali)-[~/Desktop/apple]
$ gcc hdoc19.c -o hdoc19
(kali@kali)-[~/Desktop/apple]
$

```

Screenshot 2 - Program Execution and Output

```
(kali㉿kali)-[~/Desktop/apple]
$ ./hdoc19
Enter Global ID: 590033211
Enter number of semesters: 1

Enter number of courses in semester 1: 3
Enter course code for course 1: 101
Enter credit: 4
Enter grade (O/A/B/C/D/E/P): O
Enter course code for course 2: 102
Enter credit: 3
Enter grade (O/A/B/C/D/E/P): A
Enter course code for course 3: 103
Enter credit: 3
Enter grade (O/A/B/C/D/E/P): B
Semester 1 SGPA = 9.10

Global ID = 590033211
Number of Semesters = 1
Cumulative Credits = 10
CGPA = 9.10
```

Result

The C program for semester-wise SGPA and cumulative CGPA calculation was implemented with input validation. It calculates grade points according to the given conversion scale, computes each semester SGPA, and displays the final cumulative credits and CGPA.